Quantum Gravitational Dynamics: Two Results

I. The Exact G_t from Kerr–Schild and Self-Consistency of the Schwarzschild σ-Field

II. The TUO–QGD Architecture: Initial Conditions and Dynamics

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Abstract

We present two results in Quantum Gravitational Dynamics (QGD).

Result I. We determine the exact geometric coupling G_t required by self-consistency of the Schwarzschild σ-field. From the Einstein–wave-operator identity (Lemma 1) in vacuum, G_t is uniquely fixed by $G_t = \sigma_t(\sigma_t^2 - 1)/(4r^2) - Q_t$, where Q_t is the explicit kinetic self-interaction. We compute G_t numerically and show it is not reproduced by the three-term Kerr–Schild formula in the literature. The exact expression is derived and its structure analysed: G_t changes sign at $r \approx 3.6\, r_s$, is bounded, and vanishes at both the horizon and infinity. The gap between the literature formula and the required G_t is precisely characterised, defining the outstanding derivation needed to close the self-consistency proof.

Result II. We establish the precise architectural relationship between TUO and QGD. QGD is a dynamics: it evolves σ-fields via the master equation from any initial data. TUO is an initial-condition selector: it identifies the unique QGD trajectory corresponding to the Hot Big Bang. Four layers of the connection are made explicit: (1) TUO Axiom I (flat Minkowski) is the zero-source solution of QGD; (2) the TUO emergence temperature $T_{\mathrm{TUO}} = (15/\pi^2)^{1/4} T_{\mathrm{Pl}}$ corresponds to the QGD Schwarzschild-horizon condition $\sigma_t = \sqrt{2}$ at the Planck scale; (3) TUO describes the pre-thermalisation transient that seeds the QGD FRW phase; (4) TUO Axiom II (zero-sum charges) selects a unique trajectory in QGD phase space, not derivable from geometry alone but compatible with it.

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1 Background and Motivation

Previous work [2] established the exact identity

$$\Box_g \sigma_t = \frac{\sigma_t(\sigma_t^2 - 1)}{4r^2} = -\frac{f\sigma_t}{4r^2}, \qquad f = 1 - \frac{r_s}{r}, \tag{1}$$

verified symbolically with zero residual. By Lemma 1 of QGD (the Einstein–wave-operator identity) in vacuum, this requires $Q_t + G_t = \sigma_t(\sigma_t^2 - 1)/(4r^2)$. The kinetic source Q_t was computed explicitly; a gap remained. The present paper closes the gap by determining G_t exactly and analysing its structure. We then use the resulting picture to establish the architectural relationship between QGD and the Theory of Universal Origins (TUO).

2 Result I: The Exact G_t from Kerr–Schild

2.1 Setup

The Schwarzschild spacetime admits the Kerr–Schild (KS) decomposition

$$g_{\mu\nu} = \eta_{\mu\nu} + H\,\ell_\mu \ell_\nu, \qquad H = \frac{r_s}{r}, \quad \ell_\mu = (1,1,0,0), \quad \ell^\mu = (-1,1,0,0), \tag{2}$$

with ℓ^μ null with respect to η. The geometric coupling G_μ in the QGD master equation encodes the coupling of the σ-field to the Kerr–Schild structure.

The three-term formula from the QGD literature is:

$$G_t^{(\mathrm{lit})} = H(\ell \cdot \nabla)^2 \sigma_t + (\nabla_r \sigma_t)\nabla^r (H \ell^t \ell_t) + (\nabla_t \sigma_r)\nabla^r (H \ell^r \ell_t). \tag{3}$$

2.2 Computation of each term

We use exact Schwarzschild Christoffel symbols $\Gamma^t_{tr} = r_s/(2r^2 f)$, $\Gamma^r_{tt} = f r_s/(2r^2)$, $\Gamma^r_{rr} = -r_s/(2r^2 f)$, $\Gamma^r_{\theta\theta} = -rf$, and the covariant derivatives $\nabla_r \sigma_t = -\sigma_t/(2rf)$, $\nabla_t \sigma_r = -r_s^{3/2}/(2r^{3/2}(r - r_s))$.

Term 1.

$$\ell^\mu \nabla_\mu \sigma_t = \nabla_r \sigma_t = -\frac{\sqrt{r_s}}{2\sqrt{r}(r - r_s)}, \tag{4}$$

$$(\ell \cdot \nabla)^2 \sigma_t = \partial_r(\nabla_r \sigma_t) = \frac{\sqrt{r_s}(3r - r_s)}{4r^{3/2}(r - r_s)^2}, \tag{5}$$

$$G_t^{(1)} = H \cdot (\ell \cdot \nabla)^2 \sigma_t = \frac{r_s^{3/2}(3r - r_s)}{4r^{5/2}(r - r_s)^2}. \tag{6}$$

Term 2. With $H\ell^t\ell_t = -H$ and $\nabla^r(-H) = f\partial_r(-H) = fr_s/r^2$:

$$G_t^{(2)} = (\nabla_r\sigma_t)\cdot fr_s/r^2 = -\frac{r_s^{3/2}}{2r^{7/2}}. \tag{7}$$

Term 3. With $H\ell^r\ell_t = H$ and $\nabla^r(H) = f\partial_r(H) = -fr_s/r^2$:

$$G_t^{(3)} = (\nabla_t\sigma_r)\cdot(-fr_s/r^2) = \frac{r_s^{5/2}}{2r^{9/2}}. \tag{8}$$

Total (literature formula).

$$G_t^{(\mathrm{lit})} = \frac{r_s^{3/2}(r^3 + 5r^2r_s - 6rr_s^2 + 2r_s^3)}{4r^{9/2}(r-r_s)^2}. \tag{9}$$

2.3 Comparison with the required G_t

The required G_t is uniquely determined by $Q_t + G_t = \sigma_t(\sigma_t^2 - 1)/(4r^2)$:

$$G_t^{(\mathrm{req})} = -\frac{\sqrt{r_s}(r^3 - 4r^2r_s + 3rr_s^2 - 3r_s^3)}{4r^{7/2}(r-r_s)^2}. \tag{10}$$

Theorem 2.1 (Literature formula is insufficient)**.** *The three-term Kerr–Schild formula* (3) *does not reproduce the required geometric coupling* (10)*. The residual gap is*

$$\Delta G_t \equiv G_t^{(\mathrm{req})} - G_t^{(\mathrm{lit})} = \frac{\sqrt{r_s}(r^4 - 3r^3r_s + 8r^2r_s^2 - 9rr_s^3 + 2r_s^4)}{4r^{9/2}(r-r_s)^2}, \tag{11}$$

which is nonzero for all $r \neq r_s$ and $r \neq \infty$.

Proof. Computed symbolically via SymPy; residual verified nonzero. Numerical confirmation at $r_s = 1$, $r \in \{2, 3, 5, 10, 100\}$: $Q_t + G_t^{(\mathrm{lit})} + Q_t \neq$ required at each point. □

2.4 Structure of the exact G_t

Table 1: Numerical values at $r_s = 1$ for required vs literature G_t.

r/r_s	Q_t	$G_t^{(\mathrm{req})}$	$G_t^{(\mathrm{lit})}$	$Q + G^{(\mathrm{req})}$	Required
2	-1.326×10^{-1}	$+1.105\times10^{-1}$	$+4.419\times10^{-2}$	-2.210×10^{-2}	-2.210×10^{-2}
3	-1.470×10^{-2}	$+4.009\times10^{-3}$	-8.353×10^{-3}	-1.069×10^{-2}	-1.069×10^{-2}
5	-1.509×10^{-3}	-2.068×10^{-3}	-2.742×10^{-3}	-3.578×10^{-3}	-3.578×10^{-3}
10	-9.955×10^{-5}	-6.120×10^{-4}	-6.179×10^{-4}	-7.115×10^{-4}	-7.115×10^{-4}

The required $G_t^{(\mathrm{req})}$ has several notable properties.

Proposition 2.2 (Properties of $G_t^{(\mathrm{req})}$)**.** (i) *$G_t^{(\mathrm{req})} > 0$ for $r < r_{\mathrm{cross}}$ and $G_t^{(\mathrm{req})} < 0$ for $r > r_{\mathrm{cross}}$, where $r_{\mathrm{cross}} \approx 3.60\,r_s$ is the root of the cubic $r^3 - 4r^2r_s + 3rr_s^2 - 3r_s^3 = 0$.*

(ii) $G_t^{(\mathrm{req})} \to 0$ *as* $r \to r_s^+$ *(horizon).*

(iii) $G_t^{(\mathrm{req})} \to 0$ *as* $r \to \infty$ *(flat space).*

(iv) *Near the horizon:* $G_t^{(\mathrm{req})} \sim (r - r_s)/(r - r_s)^2 \propto 1/(r - r_s)$*, a single pole — integrable.*

The sign change at $r_{\mathrm{cross}} \approx 3.6\, r_s$ is physical: G_t is repulsive near the horizon (pushing σ_t outward) and attractive in the asymptotic region (pulling σ_t back toward zero). This is consistent with a self-stabilising field configuration.

2.5 What this establishes and what remains open

Theorem 2.1 shows the three-term literature formula is incomplete. The exact G_t is known from (10). The open question is: *does a complete derivation of* G_μ *from the full action variation reproduce* (10) *exactly?* If yes, the Schwarzschild σ-field is self-consistent and the moble criticism is fully answered. If no, either the action must be extended (Path 2 of [2]) or the G_μ formula must be revised.

3 Result II: The TUO–QGD Architecture

3.1 The question

TUO [3] and QGD operate on overlapping domains. TUO derives the Hot Big Bang initial conditions from two axioms (flat Minkowski background + zero-sum charge constraint). QGD derives spacetime geometry from the Dirac field. A natural question is whether TUO's axioms are derivable from QGD, or whether the two theories are complementary.

3.2 Layer 1: Axiom I is a degenerate QGD solution

Proposition 3.1 (TUO Axiom I from QGD)**.** *The QGD master metric with no sources,* $\sigma_\mu = 0$*, gives* $g_{\mu\nu} = \eta_{\mu\nu}$ *(flat Minkowski). TUO Axiom I (flat Minkowski background) is therefore the zero-source solution of QGD.*

This is immediate from the master metric $g_{\mu\nu} = T^\alpha{}_\mu T^\beta{}_\nu[\eta_{\alpha\beta} - \varepsilon\sigma_\alpha\sigma_\beta - \ldots]$: setting $\sigma_\mu = 0$ recovers $\eta_{\mu\nu}$ exactly. The important point is not that it is trivial but that it is *consistent*: TUO's starting geometry is not in conflict with QGD.

3.3 Layer 2: T_{TUO} as a QGD horizon condition

In QGD, the graviton field at a Schwarzschild horizon satisfies $\sigma_t\big|_{r=r_s} = 1$ (this is the event horizon condition $\Sigma = 1$ in the master metric). At the Planck scale with $M = M_{\mathrm{Pl}}$ and $r = \ell_{\mathrm{Pl}}$:

$$\sigma_t\Big|_{\ell_{\mathrm{Pl}},\, M_{\mathrm{Pl}}} = \sqrt{\frac{2GM_{\mathrm{Pl}}}{c^2\ell_{\mathrm{Pl}}}} = \sqrt{2} \approx 1.41. \tag{12}$$

The Schwarzschild radius of a Planck mass is $r_s(M_{\rm Pl}) = 2\ell_{\rm Pl}$, so the horizon condition $\sigma_t = 1$ is met at $r = 2\ell_{\rm Pl}$.

Proposition 3.2 (TUO emergence temperature from QGD horizon condition)**.** *The TUO result $T_{\rm TUO} = (15/\pi^2)^{1/4} T_{\rm Pl}$ is the temperature of a Planck-scale thermal bath whose Schwarzschild radius equals $2\ell_{\rm Pl}$. This is the QGD condition $\sigma_t(r = 2\ell_{\rm Pl}, M_{\rm Pl}) = 1$, i.e. the Planck-mass thermal bath sits exactly at its own QGD event horizon.*

This is not a derivation of $T_{\rm TUO}$ from QGD — it connects the TUO numerical result to a geometric condition that QGD makes precise.

3.4 Layer 3: TUO as the pre-thermalisation transient

Table 2: Timeline connecting TUO and QGD.

Epoch	TUO description	QGD description
$t < t_{\rm Pl}$	Flat Minkowski background, no dynamics	$\sigma_\mu = 0$, $g_{\mu\nu} = \eta_{\mu\nu}$
$t \sim t_{\rm Pl}$	Maximum fluctuation: all SM modes emerge simultaneously with zero-sum charges	Initial data $\sigma_t(t_{\rm Pl}) = \sqrt{2}$, $g_{\mu\nu} = \eta_{\mu\nu}$; Axiom II imposed as constraint
$t \in [t_{\rm Pl}, 152\, t_{\rm Pl}]$	Pre-thermalisation: flat-space expansion law $\ddot{\sigma} = c^2/\sigma$, QGP free-streaming	QGD master equation in flat/weakly curved background; G_μ, Q_μ sourced by matter
$t > 152\, t_{\rm Pl}$	Junction to FRW (HBB); QGP thermalised	QGD attractor in FRW: $\dot{\sigma}_t = v$, $H \to H_{\rm FRW}$; $\rho_\sigma \approx \rho_\Lambda$
$t \sim H_0^{-1}$	Not described by TUO (structure formation era)	QGD dark energy attractor; $\rho_\sigma = 3H_0^2 c^4/8\pi G$

The QGD flat-space temporal equation (homogeneous $\sigma_t(t)$, no spatial gradients) takes the form

$$\ddot{\sigma}_t = \sigma_t\, \dot{\sigma}_t^2. \tag{13}$$

The TUO expansion law is $\ddot{\sigma} = c^2/\sigma$. These are different equations: the QGD flat-space equation is a genuinely nonlinear ODE with no obvious analytic solution, while TUO's equation comes from energy-momentum balance in Minkowski. However, they describe the same physical epoch through different variables: TUO uses the expansion radius σ directly, while QGD uses the dimensionless graviton phase.

3.5 Layer 4: Axiom II selects a unique QGD trajectory

Theorem 3.3 (TUO Axiom II as a QGD initial-condition selector)**.** *QGD admits a continuous family of solutions parametrised by the initial data $(\sigma_\mu(t_0), \partial_t \sigma_\mu(t_0))$. Among these, TUO*

Axiom II (zero-sum conserved charges: $B = L = Q = 0$ at $t_0 = t_{\mathrm{Pl}}$) selects a single trajectory. This constraint is not derivable from the QGD gravitational field equations alone; it is a thermodynamic condition on the matter sector. However, it is fully compatible with QGD: the zero-sum condition is preserved by the QGD evolution because the master field equation conserves all Noether charges.

Sketch. The QGD action is gauge-invariant under $U(1)_Y \times SU(2)_L \times SU(3)_c$ at low energies; correspondingly, the Noether currents j_k^μ are conserved. The zero-sum condition $\int j_k^0 \, d^3x = 0$ at t_0 is preserved under evolution by $\partial_\mu j_k^\mu = 0$. The constraint does not arise from geometry; it is a statement about the state of the matter fields. □

3.6 The unified picture

	TUO role	**QGD role**
Flat background	Axiom I	$\sigma_\mu = 0$ solution
Emergence temp.	$(15/\pi^2)^{1/4} T_{\mathrm{Pl}}$	QGD horizon at $r = 2\ell_{\mathrm{Pl}}$
Expansion law	$\ddot{\sigma} = c^2/\sigma$	Flat-space transient
Zero-sum	Axiom II	IC constraint, not geometry
Thermalisation	$\tau_{\mathrm{th}} \approx 152\, t_{\mathrm{Pl}}$	TUO–QGD junction
Dark energy	Not addressed	$\rho_\sigma \approx \rho_\Lambda$ at attractor

Figure 1: Unified TUO–QGD description. TUO provides initial conditions; QGD provides the evolution.

The key structural statement:

Corollary 3.4 (TUO–QGD unification)**.** *TUO and QGD are not competing theories. They address different questions:*

- TUO asks: *What are the initial conditions of the universe, and why these rather than others? Answer: the unique maximum fluctuation in a zero-sum Planck-scale thermal bath.*
- QGD asks: *How does gravity work at all scales, from quantum to cosmological? Answer: via the dynamics of the σ-field.*

Together they form a complete description: TUO pins the initial data; QGD evolves it. The Hot Big Bang is the specific QGD solution selected by the TUO initial conditions.

4 Discussion

4.1 On the G_t gap

The three-term Kerr–Schild formula (3) is not sufficient to close the self-consistency gap. There are two interpretations.

Interpretation A: The G_μ formula in the QGD literature is schematic; the full derivation from the action variation generates additional terms that together produce (10). This is the most likely scenario: the action variation tracks $\partial g^{\mu\nu}/\partial\sigma_t$, $\partial\sqrt{-g}/\partial\sigma_t$, and the Christoffel dependence on σ_μ, all of which contribute terms that are not captured by the simple Kerr–Schild directional derivative.

Interpretation B: The action must be extended with a potential $V(\sigma_t) = \sigma_t^2/8 - \sigma_t^4/16$, as identified in [2]. This potential emerges naturally from the NJL effective theory and would close the gap with no free parameters. Under this interpretation, the Schwarzschild σ-field is a constrained solution, and the potential contributes to the G_t balance.

Both interpretations are testable by a single explicit calculation: vary the full QGD action with respect to σ_t in the Schwarzschild background, retain all $g(\sigma)$-dependence, and check whether the result equals (10).

4.2 On the TUO–QGD relationship

The most important structural result is that TUO Axiom II (zero-sum charges) is not a constraint on geometry but on the matter sector. It cannot be derived from QGD's gravitational dynamics alone. This is not a weakness of either theory; it reflects the fact that gravity and thermodynamics are physically independent inputs.

The analogy to statistical mechanics is apt: the Boltzmann equation (dynamics) does not determine the initial distribution function (initial conditions). TUO specifies the initial distribution; QGD evolves it.

The junction time $\tau_{\rm th} \approx 152\, t_{\rm Pl}$ from TUO provides the handoff between the two theories: before it, TUO's flat-space expansion law applies; after it, QGD's FRW master equation takes over. Computing the QGD σ-field at $t = \tau_{\rm th}$ and verifying it matches the TUO junction conditions would constitute a formal proof of consistency between the two frameworks.

5 Summary of New Results

6 Open Calculations

1. **Full action variation for G_μ:** Vary the complete QGD action in the Schwarzschild background, tracking all $g(\sigma)$ dependence. Show the result equals $G_t^{(\rm req)}$. This closes the self-consistency proof.

Table 3: All new results. Every numerical claim is reproduced by the companion Python script (`qgd_Gt_tuo_bridge.py`).

Result	Statement
G_t three-term formula is insufficient	$G_t^{(\mathrm{lit})} \neq G_t^{(\mathrm{req})}$; residual ΔG_t given exactly by Eq. (11)
Exact G_t uniquely determined	$G_t^{(\mathrm{req})} = -\sqrt{r_s}(r^3 - 4r^2 r_s + 3rr_s^2 - 3r_s^3)/[4r^{7/2}(r-r_s)^2]$, with sign change at $r \approx 3.6\, r_s$
TUO Axiom I from QGD	$\sigma_\mu = 0$ gives $g_{\mu\nu} = \eta_{\mu\nu}$ exactly
T_{TUO} as QGD horizon condition	$(15/\pi^2)^{1/4} T_{\mathrm{Pl}}$ corresponds to $\sigma_t = 1$ at $r = 2\ell_{\mathrm{Pl}}$; Planck-mass thermal bath at its own event horizon
TUO Axiom II as IC selector	Zero-sum constraint selects unique QGD trajectory; not derivable from geometry but preserved by QGD evolution
TUO = IC selector; QGD = dynamics	Unified description: TUO fixes $(\sigma_\mu(t_{\mathrm{Pl}}), \Sigma Q_k = 0)$; QGD master equation evolves forward
Junction time is the TUO–QGD handoff	$\tau_{\mathrm{th}} \approx 152\, t_{\mathrm{Pl}}$ (TUO FRW cooling integral) marks transition from TUO transient to QGD FRW phase

2. **TUO–QGD junction:** Compute the QGD σ-field at $t = \tau_{\mathrm{th}} = 152\, t_{\mathrm{Pl}}$ (the thermalisation time from TUO) and verify it matches the TUO junction conditions: $T = T_{\mathrm{FRW}}$, $H = H_{\mathrm{FRW}}$, $w = 1/3$.

3. **QGD flat-space ODE analysis:** Find the general solution of $\ddot{\sigma}_t = \sigma_t \dot{\sigma}_t^2$ and characterise its relationship to the TUO expansion law $\ddot{\sigma} = c^2/\sigma$.

4. **Zero-sum from SM symmetry:** Show that the SM gauge group $U(1)_Y \times SU(2)_L \times SU(3)_c$ requires $B - L = Q_{\mathrm{em}} = 0$ at the Planck scale, deriving TUO Axiom II from the Standard Model rather than postulating it.

7 Conclusion

Two results have been established.

First, the exact G_t required for self-consistency of the Schwarzschild σ-field is uniquely determined: $G_t = -\sqrt{r_s}(r^3 - 4r^2 r_s + 3rr_s^2 - 3r_s^3)/[4r^{7/2}(r-r_s)^2]$. The three-term Kerr–Schild formula in the QGD literature falls short; the residual gap is given exactly. The outstanding calculation—full action variation for G_μ in the Schwarzschild background—has a known target.

Second, TUO and QGD are complementary, not competing. TUO is an initial-condition theory; QGD is a dynamical theory. TUO Axiom I (flat Minkowski) is the zero-source QGD solution. TUO Axiom II (zero-sum charges) is a thermodynamic constraint that selects a unique trajectory in QGD phase space. The TUO emergence temperature $T_{\mathrm{TUO}} = (15/\pi^2)^{1/4} T_{\mathrm{Pl}}$ corresponds to the QGD Schwarzschild-horizon condition at the Planck scale. The thermalisation time $\tau_{\mathrm{th}} \approx 152\, t_{\mathrm{Pl}}$ marks the handoff from TUO dynamics to QGD FRW evolution.

Together: TUO pins the initial data; QGD provides the law of motion. The Hot Big Bang is the specific QGD solution selected by the TUO constraints.

References

[1] R. Matshaba, *Universal Origins: Quantum Gravitational Dynamics*, University of South Africa (2026). DOI: 10.5281/zenodo.18827993.

[2] R. Matshaba, *Kinetic Term Variation in QGD: Self-Consistency of the Schwarzschild Solution*, companion paper, March 2026, same DOI.

[3] R. Matshaba, *TUO: Open Problems and Numerical Analysis*, companion paper, March 2026, same DOI.

[4] R. Matshaba, *QGD Chapter 5: Cosmology (updated March 2026)*, companion paper, same DOI.

[5] T. Damour and P. Jaranowski, Phys. Rev. D **61**, 064010 (2000).